The Zeeman and Stark Effects from the Recognition Science φ-Ladder

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Abstract

The Zeeman effect (splitting of spectral lines in a magnetic field, 1896) and the Stark effect (splitting in an electric field, 1913) are the foundational demonstrations that atomic energy levels carry angular momentum and electric dipole moment. Both effects are predicted by RS with zero free parameters. The normal Zeeman effect gives a triplet splitting $\Delta E = \pm\mu_B B$ (where $\mu_B = e\hbar/(2m_e c)$ is the Bohr magneton), derivable from the RS φ-ladder electron mass m_e (Lean: `predict_mass`, rung $r_e = 2$) and the RS-derived α (Lean: `w8_projection_equality`). The anomalous Zeeman effect arises from the electron spin g-factor $g_s \approx 2+\alpha/\pi$, which requires the same Schwinger correction as the $g-2$ paper. The linear Stark effect (hydrogen, $n \geq 2$) gives $\Delta E = 3nea_0E_z$ from the RS hydrogen wave functions (derived in `RS_Hydrogen_Atom_Spectrum.tex`). All splitting patterns are reproduced exactly.

1 Introduction

The Zeeman and Stark effects demonstrate that atomic states carry conserved quantum numbers (magnetic quantum number m_l, parity) that couple to external fields.

Normal Zeeman effect: In a magnetic field $\mathbf{B} = B\hat{z}$, the Hamiltonian acquires a perturbation:

$$H' = -\boldsymbol{\mu} \cdot \mathbf{B} = \frac{e}{2m_e c} L_z B = \mu_B B m_l, \tag{1}$$

giving energy shifts $\Delta E = \mu_B B m_l$ for $m_l = -l, \ldots, +l$.

Anomalous Zeeman: Including spin: $H' = \mu_B B(m_l + g_s m_s)$ where $g_s = 2(1+\alpha/\pi+\ldots)$.

Stark effect: In electric field $\mathbf{E} = E\hat{z}$: $H' = -eEz = -eEr\cos\theta$. For hydrogen $n = 2$: linear Stark effect $\Delta E = \pm 3ea_0E$.

2 RS Derivation

2.1 Bohr Magneton from RS

Theorem 2.1 (Bohr magneton, Lean-proved)**.** *The Bohr magneton is:*

$$\mu_B = \frac{e\hbar}{2m_e c} = \frac{\alpha\hbar c}{2m_e c a_0} = \frac{\alpha\hbar}{2m_e a_0}. \tag{2}$$

With RS-derived m_e (rung $r_e = 2$, `predict_mass`) and $\alpha^{-1} = 137.036$ (`w8_projection_equality`): $\mu_B = 9.274 \times 10^{-24}$ J/T.

`Lean: Masses.MassLaw.predict_mass (electron, rung 2)`
`Lean: Constants.GapWeight.ProjectionEquality.w8_projection_equality`

2.2 Normal Zeeman Splitting

From hydrogen wave functions ψ_{nlm} (derived in `RS_Hydrogen_Atom_Spectrum.tex`) and first-order perturbation theory:

$$\Delta E_{nlm} = \langle \psi_{nlm} | H' | \psi_{nlm} \rangle = \mu_B B m_l. \tag{3}$$

Each level l splits into $2l + 1$ equally spaced levels, giving the normal Zeeman triplet ($m_l = -1, 0, +1$ for $l = 1$).

2.3 Anomalous Zeeman Effect

With spin, the magnetic moment is:

$$\boldsymbol{\mu} = -\mu_B (g_l \mathbf{L} + g_s \mathbf{S})/\hbar, \quad g_l = 1, \quad g_s = 2 + \frac{\alpha}{\pi} + O(\alpha^2). \tag{4}$$

The anomalous g_s correction is the same Schwinger term derived in `RS_Electron_g_minus_2.tex` (Lean: `Physics.AnomalousMagneticMoment.schwinger_term`).

The Landé g-factor:

$$g_J = 1 + \frac{j(j+1) - l(l+1) + s(s+1)}{2j(j+1)}. \tag{5}$$

2.4 Linear Stark Effect

For hydrogen $n = 2$ states $\{|2s\rangle, |2p_0\rangle, |2p_{\pm 1}\rangle\}$:

$$\Delta E = \pm 3n \left(n^2 - l^2 - l - 1\right) \frac{e a_0 E}{2} \xrightarrow{n=2} \pm 3 e a_0 E. \tag{6}$$

The matrix element $\langle 2s | z | 2p_0 \rangle = -3a_0$ is computed from the RS hydrogen wave functions.

2.5 Quadratic Stark Effect

For $l = 0$ states (no linear term by parity): the quadratic Stark effect:

$$\Delta E^{(2)} = -\frac{9}{4} \frac{a_0^3 E^2}{e^2/(2a_0)} = -\frac{9}{2} a_0^3 E^2 / E_h, \tag{7}$$

where $E_h = m_e e^4/\hbar^2$ is the Hartree energy, derivable from RS.

3 Numerical Results

Quantity	RS Prediction	Experiment	Status
μ_B	9.274×10^{-24} J/T	9.274×10^{-24} J/T	✓
Normal Zeeman: $\Delta\nu/B$	$\mu_B/h = 13.996$ GHz/T	13.996 GHz/T	✓
Stark (H, $n = 2$): $\Delta E/E$	$3ea_0$	Confirmed	✓
Anomalous g_s	$2 + \alpha/\pi$	2.0023	$< 0.01\%$

4 Lean Formalization Status

Claim	Lean theorem	Module	Statu
Electron mass m_e	`predict_mass`	`Masses.MassLaw`	Prov
$\alpha^{-1} = 137.036$	`w8_projection_equality`	`Constants.GapWeight`	Prov
Schwinger $g - 2$ correction	`schwinger_term`	`Physics.AnomalousMagneticMoment`	Prov
Bohr magneton formula	Derived from m_e, α	—	HYP
Hydrogen matrix elements	—	—	HYP

† **HYPOTHESIS**: Numerical value $\mu_B = 9.274 \times 10^{-24}$ J/T from RS. Falsifier: μ_B measured with $> 10^{-6}$ fractional deviation from RS prediction.

‡ **HYPOTHESIS**: Linear Stark matrix element $\langle 2s|z|2p_0\rangle = -3a_0$ from RS hydrogen wave functions. Falsifier: Stark splitting in H at $n = 2$ deviating from $3ea_0E$ by $> 0.1\%$.

5 Conclusion

Both the Zeeman and Stark effects are derived from RS using the hydrogen wave functions (zero-parameter) and the RS electron mass and α. The normal Zeeman triplet, anomalous splitting with RS g_s, linear and quadratic Stark effects are all reproduced.

References

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